

CMPSCI 670: Computer Vision, Fall 2014



Key to Dreams, *René Magritte*

Instructor: [Subhansu Maji](#)

Lecture: Monday and Wednesday, 2:30 - 3:45 PM, CS 142

Office hours: Wednesday, 3:45 - 4:45 PM, CS 274

Graders: Ryan Szeto, Tung (Steven) Pham

Overview

This course will explore current techniques for the analysis of visual data (primarily color images). In the first part of the course we will examine the physics and geometry of image formation, including the design of cameras and the study of color sensing in the human eye. In each case we will look at the underlying mathematical models for these phenomena. In the second part of the course we will focus on algorithms to extract useful information from images. This includes detection of reliable interest points for applications such as image alignment, stereo and instance recognition; robust representations of images for recognition; and principles for grouping and segmentation. Time permitting we will look at some additional topics at the end of the course.

Course assignments will highlight several computer vision tasks and methods. For each task you will construct a basic system, then improve it through a cycle of error analysis and model redesign. There will also be a final project, which will investigate a single topic or application in greater depth. This course assumes a good background in basic probability, linear algebra, and ability to program in MATLAB. Prior experience in signal/image processing is useful but not required.

Textbooks (recommended)

- [Computer Vision: A Modern Approach](#) by David Forsyth and Jean Ponce (2nd ed.)
- [Computer Vision: Algorithms and Applications](#) by Richard Szeliski (pdf available [online](#))

Announcements

- **12/22/14** Grades posted.
- **12/10/14** Final report due at 11:55 pm, Monday (December 15) via email.
- **12/02/14** We will meet on Thursday (12/4) at 5:30pm, CS 142 to wrap up the project presentations. There will be pizza!
- **12/01/14** Slides for 11/26 lecture on optical flow posted (class was cancelled due to snow storm)
- **11/17/14** Guest lecture by Aditya Khosla on Wednesday on "Crafting the Perfect Selfie using Computer Vision"
- **11/17/14** Homework 4 grades posted.
- **11/11/14** Office hours this week are at Thursday 3:45-4:35, CS 274
- **11/05/14** Homework 3 grades posted.
- **11/05/14** Homework 5 posted.
- **11/03/14** Office hours this week are at Monday 3:45-4:35, CS 274
- **10/22/14** Homework 4 posted.

- **10/22/14** Office hours this week are at Thursday 2:30-3:30, CS 274
- **10/19/14** Homework 2 grades posted.
- **10/19/14** Homework 3 deadline extended to Wednesday, Oct 22.
- **10/14/14** [Final project guidelines](#) posted.
- **10/08/14** Office hours this week are at Thursday 3:45-4:35, CS 274
- **10/07/14** Homework 1 grades posted.
- **10/06/14** Homework 3 posted.
- **09/28/14** There was a bug in the initial code for HW2. Download the latest code.zip from the [homework page](#).
- **09/24/14** Homework 2 posted.
- **09/24/14** Office hours this week are at Thursday 3:45-4:35, CS 274
- **09/17/14** Instructions for HW1 submission posted on the [homework page](#).
- **09/11/14** Make sure you can log in to your edLab accounts. Homeworks will be collected and tested via edLab.
- **09/11/14** Further readings for each lecture can be found in the last slide of each lecture.
- **09/08/14** Homework 1 posted.
- **09/03/14** Office hours this week are at Thursday 3:45-4:35, CS 274
- **08/21/14** Tentative course outline posted.

Schedule

Week	Date	Topic	Links	Homework (out)	Homework (due)
1	Sep 3	Course introduction [pdf , key]	MATLAB tutorial and setup		
	Sep 8	Image formation [pdf , key]	Edlab resources	HW1: Color images	
2	Sep 10	Cameras [pdf , key]			
	Sep 15	Color [pdf , key]			
3	Sep 17	Light and shading [pdf , key]			
	Sep 22	Linear filtering [pdf , key]			HW1 due
4	Sep 24	Edge detection [pdf , key]		HW2: Photometric stereo	HW1 due
	Sep 29	Corner detection [pdf , key]			
5	Oct 1	Blob detection [pdf , key]			
	Oct 6	Texture [pdf , key]		HW3: Blob detection	HW2 due
6	Oct 8	Texture <i>continued</i> [pdf , key]			
	Oct 13		No class (Columbus day)		
7	Oct 14	Grouping [pdf , key]		FP guidelines	
	Oct 15	Grouping <i>continued</i> [pdf , key]			
	Oct 20	Alignment [pdf , key]			HW3 due
8	Oct 22	Alignment <i>continued</i> [pdf , key]		HW4: Grouping	HW3 due
9	Oct	Recognition <i>introduction</i> [pdf ,			Abstracts due

	27	key]		
	Oct	Machine learning <i>introduction</i>		
	29	[pdf , key]		
10	Nov 3	Image representation [pdf , key]		
	Nov 5	Object detection [pdf , key]	HW5: Recognition	HW4 due
	Nov	Object detection <i>continued</i>		
11	10	[pdf , key]		
	Nov	No class (post Veteran's day. Tuesday class schedule will be followed)		
	12			
	Nov	Deep learning [pdf , key]		
	17			
12	Nov	Guest lecture: Aditya Khosla , MIT	Title: Crafting the Perfect Selfie using Computer Vision [pdf , pptx]	HW5 due
	Nov	Human-centric computer vision		
	24	[pdf , key]		
13	Nov	Optical flow and tracking	Class cancelled due to snow	
	26	[pdf , key]		
14	Dec 1	Project presentations		
	Dec 3	Project presentations <i>continued</i>		
	Dec			
	13 15			Final report due

Acknowledgements

Many of the slides and homework assignments are based on excellent computer vision courses taught elsewhere by [Svetlana Lazebnik](#), [Alyosha Efros](#), [Alexander Berg](#), [Steven Seitz](#), [James Hays](#), [Charles Flowkes](#), [Kirsten Grauman](#) and many others. Many thanks to [Richard Szeliski](#) for making the textbook available online for free.